

Cambridge (CIE) IGCSE Chemistry
Paper 1: Multiple Choice (Set E)
Core

Saturday 26 April 2025

Afternoon (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

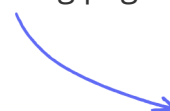
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



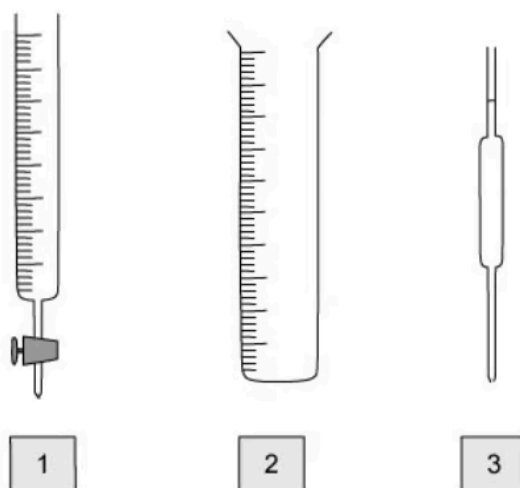
1 Four substances are shown below.

In which of these substances are the particles close together and moving slowly past each other?

- A. Ice
- B. Air
- C. Steam
- D. Water

(1 mark)

2 Measuring volumes of liquids is a fundamental aspect of experimental chemistry. The diagram shows three different pieces of apparatus used for this task.



Which row is correct?

	1	2	3
A	burette	measuring cylinder	pipette
B	burette	pipette	measuring cylinder
C	pipette	measuring cylinder	burette
D	measuring cylinder	burette	pipette

(1 mark)

3 Pure H₂O boils at 100 °C and freezes at 0 °C.

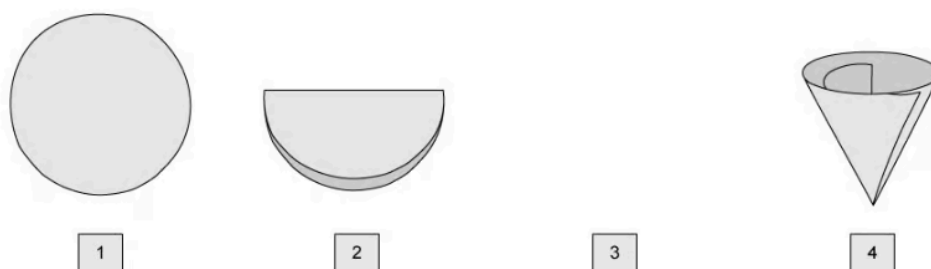
What is the melting point and boiling point of water after the addition of potassium chloride?

	Boiling point / °C	Freezing point / °C
A	103	-3
B	103	3
C	97	-3
D	97	No change

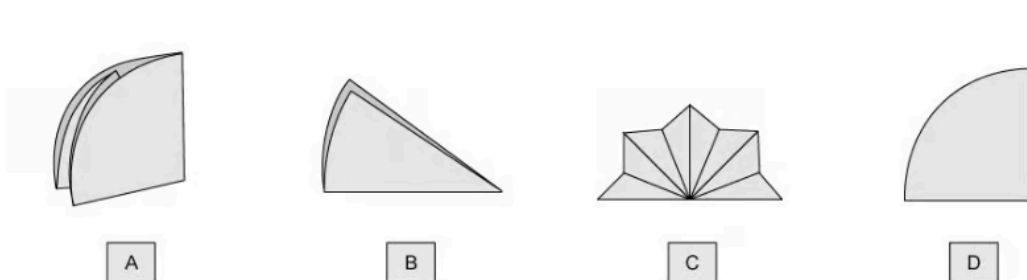
(1 mark)

4 A student has performed an experiment which produced a precipitate. She correctly chooses to separate the precipitate using the process of filtration. She prepares the filter

paper to be placed into the funnel by folding it as shown below.



Which diagram correctly illustrates the filter paper at stage 3?



(1 mark)

- 5 Most of the elements on the Periodic Table are metals. Metals can be distinguished from non-metals by differences in their physical properties.

Which row correctly describes a metal?

	Malleability	Heat & electrical conductivity
A	×	✓
B	✓	×
C	✓	✓
D	×	×

- A.** Option A
- B.** Option B
- C.** Option C
- D.** Option D

(1 mark)

6 Element X contains 4 electrons in its outer shell and 125 neutrons in its nucleus. What is element X?

- A.** Beryllium
- B.** Carbon
- C.** Tin
- D.** Lead

(1 mark)

7 CH₄, H₂O and HCl are covalent compounds.

Which atoms in these compounds do not use all of their outer shell electrons in bonding?

- A. C and O
- B. C and H
- C. Cl and O
- D. Cl and C

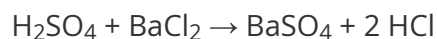
(1 mark)

8 Which statement about graphite and diamond is not correct?

- A. Graphite is a good lubricant due to the weak forces in between the layers.
- B. Diamond is used as a drill tip because it is an extremely hard substance.
- C. Graphite has a low melting point due to the weak forces in between the layers.
- D. Diamond is an allotrope of carbon.

(1 mark)

9 Dilute sulfuric acid and barium chloride solution react to produce barium sulfate and hydrochloric acid, as shown in the reaction below:



Which row represents the correct state symbols for the reaction?

	H_2SO_4	BaCl_2	BaSO_4	HCl
A	(aq)	(l)	(s)	(aq)
B	(l)	(aq)	(aq)	(l)
C	(aq)	(aq)	(s)	(aq)
D	(l)	(aq)	(l)	(aq)

(1 mark)

- 10** Solid lead(II) bromide is heated until it is molten. The molten lead (II) bromide then undergoes electrolysis.

Which substances are formed at each electrode?

	anode	cathode
A	bromine	lead
B	lead	bromine
C	bromine	hydrogen
D	hydrogen	lead

(1 mark)

- 11** Which is true about chemical energy changes?

	Reaction type	Energy flows	Temperature of surroundings
A	exothermic	to the surroundings	increases
B	exothermic	from the surroundings	decreases
C	endothermic	to the surroundings	decreases
D	endothermic	from the surroundings	increases

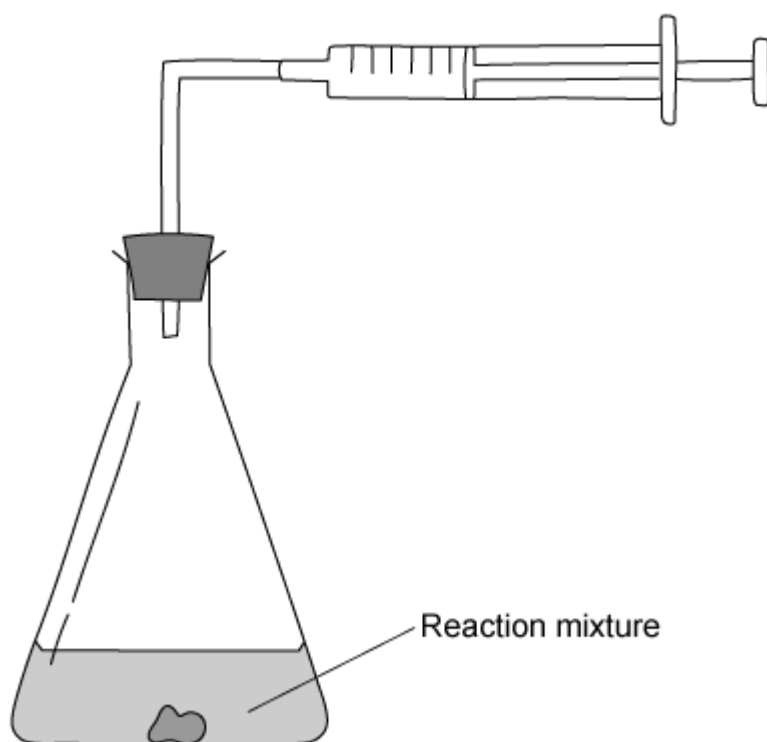
(1 mark)

12 Which of the following processes represents a physical change?

- A.** Acid base neutralisation
- B.** Metal displacement reactions
- C.** Boiling ethanol
- D.** Combustion of magnesium

(1 mark)

13 A student was measuring the rate of reaction using the apparatus in the diagram.



What reaction is the student studying?

- A.** $\text{CaO} + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O}$
- B.** $\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$
- C.** $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$
- D.** $\text{KCl} + \text{AgNO}_3 \rightarrow \text{KNO}_3 + \text{AgCl}$

(1 mark)

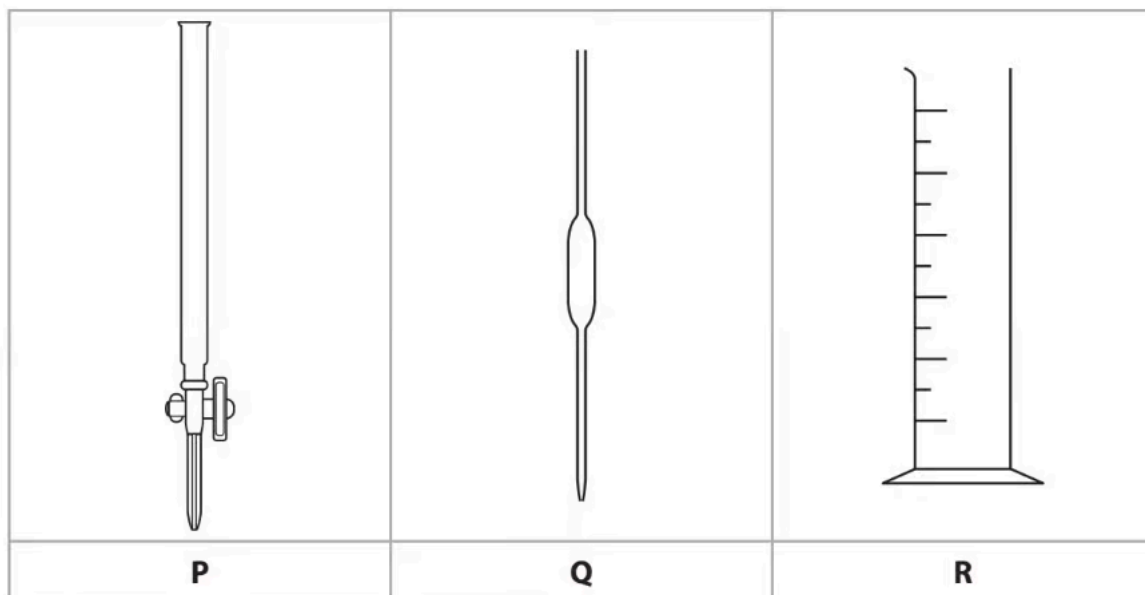
14 Which of the following statements is not correct?

- A.** Ammonia is released when a base reacts with an ammonium salt.
- B.** Neutralisation occurs when an acid and an alkali react together.
- C.** Carbon dioxide is released when an acid and a carbonate react together.
- D.** The higher the pH of a solution, the higher its acidity.

(1 mark)

15 Different salts can be prepared by titrating an alkali with an aqueous acid.

Three pieces of apparatus, **P**, **Q** and **R**, used to measure volumes of liquid are shown below.

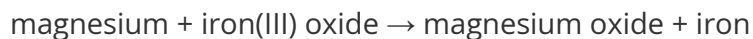


What are the names of the three pieces of apparatus, **P**, **Q** and **R**?

	P	Q	R
A	pipette	burette	measuring cylinder
B	burette	pipette	conical flask
C	pipette	volumetric flask	burette
D	burette	pipette	measuring cylinder

(1 mark)

16 Magnesium reacts with iron(III) oxide in the following reaction.



Which statement about this reaction is correct?

- A.** Magnesium oxide is reduced
- B.** Iron is oxidised
- C.** Magnesium is oxidised
- D.** Iron(III) oxide is oxidised

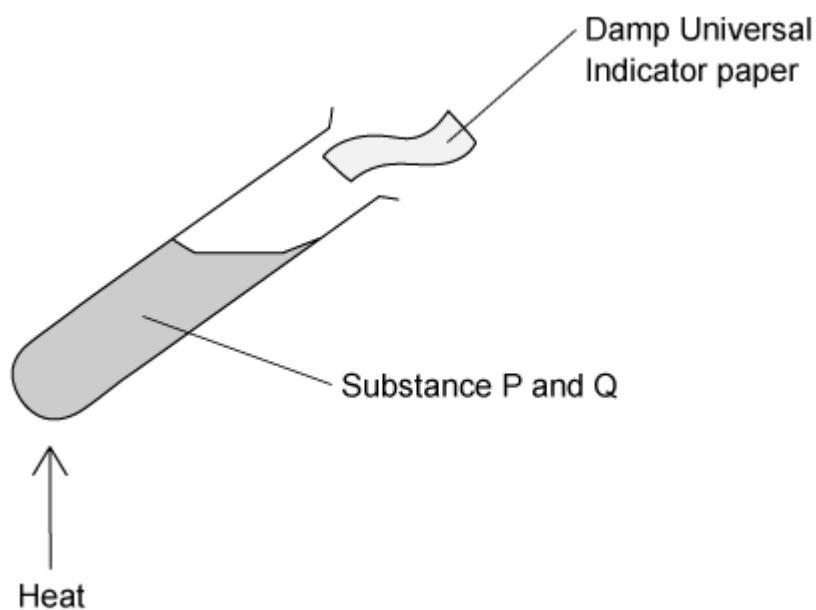
(1 mark)

17 Which reaction would result in the production of a gas that would produce a squeaky pop with a lit splint?

- A.** Iron(II) oxide + sulfuric acid
- B.** Methane + oxygen during complete combustion
- C.** Magnesium + hydrochloric acid
- D.** Carbon dioxide and water in the presence of light and chlorophyll

(1 mark)

18 A student is testing two substances, **P** and **Q**, as shown.



The results of the test are that the Universal indicator paper turns blue. What are substances **P** and **Q**?

- A.** HCl and NH_4NO_3
- B.** NaOH and NH_4NO_3
- C.** NaCO_3 and HCl
- D.** NaCO_3 and NaOH

(1 mark)

19 When an aqueous salt is crystallised, some water can be retained within the structure of the salt.

What is the name given to the type of salt that forms?

A	anhydrous salt
B	dehydrated salt
C	hydrated salt
D	soluble salt

(1 mark)

20 Which entry in the table matches the transition elements?

	density	melting point	from coloured compounds
A	high	low	no
B	high	high	yes
C	low	low	no
D	low	high	yes

(1 mark)

21 Element **X** is found in Group 7 of the Periodic Table.

Which properties are typical of element **X**?

1	Poor conductor of electricity
2	Malleable
3	High melting points
4	Form acidic oxides

A. 1 and 2

B. 1 and 4

C. 2 and 3

D. 3 and 4

(1 mark)

- 22** The elements of the Periodic table can be broadly classified into two main characters which are metals and nonmetals. Which statements are correct?

1	Magnesium is more metallic than silicon.
2	Boron is more metallic than lithium.
3	Carbon is less metallic than beryllium.
4	Sodium is less metallic than aluminium.

A. 1 and 3

B. 1 and 2

C. 2, 3 and 4

D. 1, 2 and 4

(1 mark)

23 Element X is an unreactive non-metal element. Where could this element be found on the Periodic table?

A. At the far left hand side

B. At the far right hand side

C. At the top of Group VI

D. At the bottom of Group VII

(1 mark)

24 Which of the following observations does not apply to Group I elements?

A. They are relatively soft metals.

B. They corrode quickly when exposed to air.

C. They react very quickly with water releasing hydrogen gas

D. They produce acidic solutions in reaction with water.

(1 mark)

25 Which row in the table below shows how the metal will react?

	metal	reacts with dilute acid	reacts rapidly with cold water	reacts with steam
A	copper	✓	×	×
B	calcium	×	✓	✓
C	zinc	✓	×	×
D	magnesium	✓	×	✓

- A.** Option A
- B.** Option B
- C.** Option C
- D.** Option D

(1 mark)

26 Many metals can be extracted from their ores by heating the metal oxide with carbon.

Which of these metals can't be extracted using carbon?

- A.** Zinc
- B.** Copper
- C.** Aluminium
- D.** Iron

(1 mark)

27 Properties of four elements are shown below.

Which of these elements is a metal?

	electrical conductivity when solid	electrical conductivity when liquid	melting point °C
A	low	low	-7
B	high	low	801
C	high	high	1535
D	low	low	3550

- A.** Option A

B. Option B

C. Option C

D. Option D

(1 mark)

28 Brass is an alloy of two metals

Which is the correct statement?

A. The alloy will completely dissolve in dilute hydrochloric acid.

B. Brass is formed by a chemical reaction between two metals

C. Brass can be represented by a chemical formula

D. One of the metals in the alloy will dissolve in dilute hydrochloric acid

(1 mark)

29 Human activities increase the level of nitrogen oxides, sulfur dioxide and carbon dioxide in our atmosphere.

Which row correctly shows the effect of each of these gases in our atmosphere?

	Oxides of nitrogen	Sulfur dioxide	Carbon dioxide
A	global warming	respiratory problems	photochemical smog
B	acid rain	photochemical smog	respiratory problems
C	photochemical smog	acid rain	global warming
D	respiratory problems	global warming	acid rain

(1 mark)

- 30** Reversible reactions occur when products formed in the forward reaction behave as reactants and proceed in the reverse direction to produce the original reactants again.

Which of the following statements is correct for a reversible reaction?

1	If the forward reaction is exothermic, then the reverse reaction must be endothermic
2	If the forward reaction is endothermic, then the reverse reaction must be exothermic
3	Both directions are exothermic
4	Both directions are endothermic

- A.** 1 and 2
B. 3 and 4
C. 1 only
D. 3 only

(1 mark)

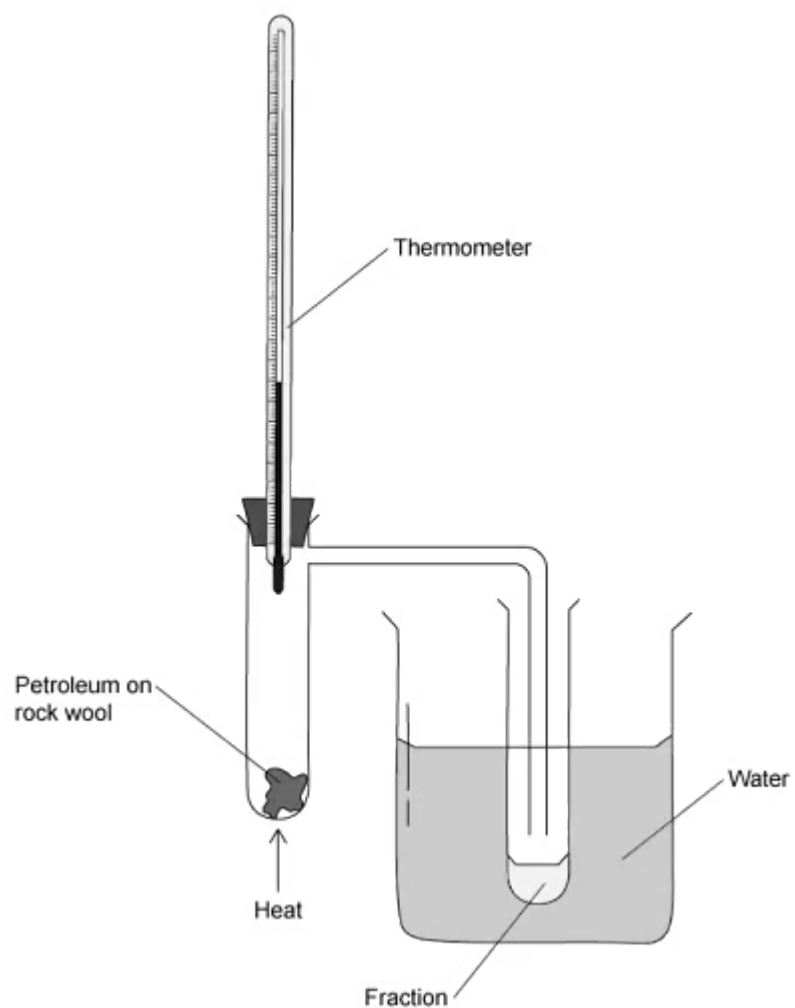
- 31** Mineral water contains dissolved salts such as magnesium chloride. Which one of the following statements about mineral water is correct?

- A.** Mineral water boils at slightly above 100 °C.
B. Mineral water is pure water.
C. Mineral water boils at exactly 100 °C.
D. Another name for mineral water is fizzy water.

(1 mark)

- 32** The diagram below shows the apparatus used to separate petroleum (crude oil) into four

fractions.

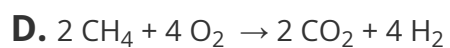
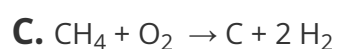
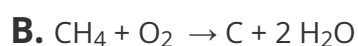
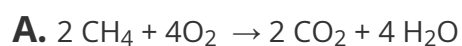


Which fraction contains the smallest hydrocarbon molecules?

fraction	boiling point range / °C
A	up to 70
B	71 to 120
C	121 to 170
D	over 171

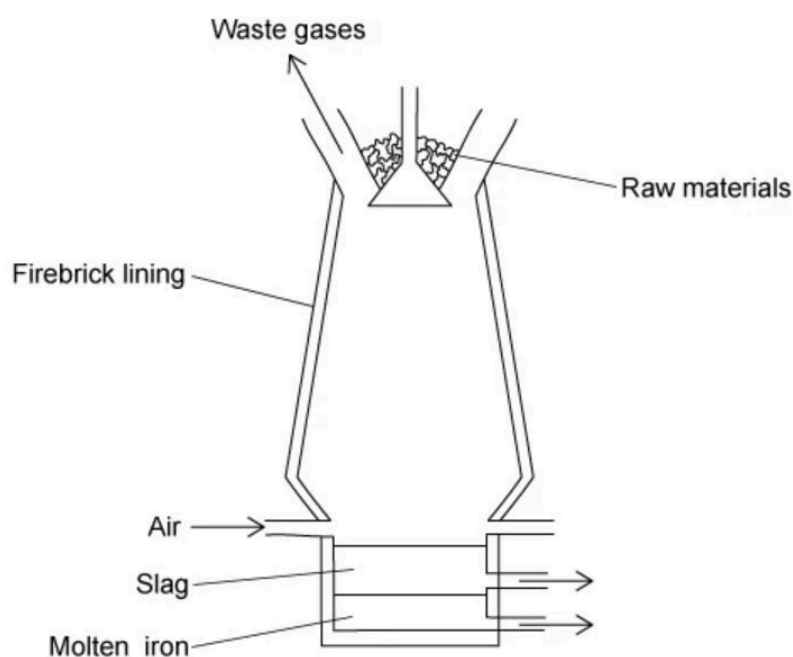
(1 mark)

33 Which of the equations show the incomplete combustion of methane?



(1 mark)

34 The blast furnace is used to extract iron from haematite.



The haematite contains silica as an impurity

What is used to react with the impurity and remove it?

- A.** Carbon
- B.** Calcium oxide
- C.** Carbon dioxide
- D.** Oxygen

(1 mark)

35 Which of the following statements about ethanol is **not** correct?

- A.** It is formed by a reaction between steam and ethane.
- B.** It can be used as a fuel.
- C.** It can be used as a solvent.
- D.** Its functional group is -OH.

(1 mark)

1	Q is unsaturated.
2	R will decolourise bromine water.
3	A molecule of P has 7 carbon atoms.

Which of the statements are correct?

- A.** 1 & 3
- B.** 1 & 2
- C.** 2 & 3
- D.** 1, 2 & 3

(1 mark)

38 Which statement about aqueous sodium hydroxide is correct?

- A.** When added to a solution containing nitrate ions and aluminium foil then warmed, the gas given off will bleach moist red litmus paper
- B.** When added to a solution containing iodide ions, a yellow precipitate is formed
- C.** When added to a solution of iron(II) ions, a green precipitate is formed which dissolves in excess sodium hydroxide to form a green solution
- D.** When added to a solution containing aluminium ions, a white precipitate forms which dissolves in excess sodium hydroxide to form a colourless solution

(1 mark)

39 Which of these statements about carboxylic acids is **not** correct?

- A.** Aqueous ethanoic acid has a pH less than 7.
- B.** Methyl orange turns yellow in aqueous ethanoic acid.
- C.** They produce carbon dioxide when reacted with a metal carbonate.
- D.** They contain the functional group -COOH.

(1 mark)

40 A macromolecule is a very large molecule.

Macromolecules can be made by joining smaller molecules together. This is called polymerisation.

Which row in the table describes the formation of a polymer?

	monomer	polymer
A	ethane	poly(ethane)
B	ethene	poly(ethene)
C	ethane	poly(ethene)
D	ethene	poly(ethane)

(1 mark)